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SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Networks **COURSE CODE:** CSA0711

COURSE OUTCOME COVERED

CO5: Measure network performance using key metrics with simulation tools

Assessment Tool Weightage: 20%

QUESTIONS: 40 **Total Marks: 40**

Annexure A

Mock Interview

QUESTIONS: 40 **Total Marks: 40**

1. Analyze how network congestion affects packet delay and packet loss. (BL4)
2. Compare the impact of star and mesh topologies on network reliability and fault tolerance. (BL4)
3. Why does increasing the number of routers between source and destination increase end-to-end delay? Explain logically. (BL4)
4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic. (BL4)
5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication. (BL5)
6. Analyze how network segmentation using VLANs improves performance and security. (BL4)
7. Compare the effects of TCP and UDP on network performance for different applications. (BL4)
8. Why is load balancing important in large enterprise networks? Analyze its impact on performance and availability. (BL5)
9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization. (BL4)
10. Evaluate the trade-off between network security measures (such as firewalls and encryption) and overall network performance. (BL5)
11. A network has high delay—how will you identify and fix the issue? (BL4)
12. If throughput is low despite high bandwidth, what could be the reasons? (BL4)
13. How would you reduce packet loss in a congested network? (BL4)
14. A VoIP call is breaking frequently—how would you troubleshoot? (BL4)
15. Given a network with high jitter, suggest improvements (BL4)
16. How will you simulate a congested network in Packet Tracer? (BL5)
17. If packets are delayed in a router, what steps will you take? (BL4)
18. How would you optimize a network for online gaming? (BL4)
19. Design a small network and explain how you will measure its performance (BL5)
20. If RTT is very high, how will you diagnose the issue? (BL4)

21. Explain network performance to a non-technical person (BL5)
22. Describe throughput using a real-life example (BL5)
23. Present how packet loss affects video streaming (BL5)
24. Explain delay types clearly in under 1 minute (BL4)
25. Describe Cisco Packet Tracer in a professional way (BL5)
26. Explain QoS in simple and technical terms (BL5)
27. How would you present your simulation results to a client? (BL4)
28. Explain jitter using a real-world analogy (BL5)
29. Describe how you tested network performance in a lab (BL4)
30. Explain the importance of performance metrics in one structured answer (BL5)
31. Introduce yourself and explain your knowledge of network performance (BL5)
32. Why are you interested in networking and simulation tools? (BL5)
33. What makes you confident in analyzing network performance? (BL5)
34. Explain a concept you are very strong in (related to networking) (BL5)
35. How would you handle a question you don't know? (BL5)
36. Describe a situation where you solved a technical problem (BL4)
37. Why should we select you for a networking role? (BL5)
38. What challenges did you face while learning Packet Tracer? (BL5)
39. How do you improve your technical skills regularly? (BL5)
40. Demonstrate confidence while explaining a complex concept simply (BL5)

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questio

MOCK INTERVIEW — MODEL ANSWERS

Section A: Analytical & Conceptual Questions (Q1–Q10)

Q1. Analyze how network congestion affects packet delay and packet loss. (BL4)

Answer: As traffic on a link approaches its capacity, packets queue up at router buffers, increasing queuing delay; once buffers overflow, incoming packets are dropped, causing packet loss. Both delay and loss rise sharply (non-linearly) near saturation, and TCP retransmissions triggered by loss add further load, which can push the network toward congestive collapse.

Q2. Compare the impact of star and mesh topologies on network reliability and fault tolerance. (BL4)

Answer: A star topology depends on a central hub/switch — if it fails, the entire network goes down, though a single node failure only isolates that node. A mesh topology provides multiple redundant paths between nodes, so a link or node failure can be routed around, giving much higher fault tolerance and reliability, at the cost of more cabling and higher deployment complexity.

Q3. Why does increasing the number of routers between source and destination increase end-to-end delay? Explain logically. (BL4)

Answer: Each router a packet passes through adds its own processing delay (examining the header and making a forwarding decision), queuing delay (waiting in the router's buffer), and transmission delay (pushing the packet onto the outgoing link), plus the propagation delay of the additional link. These per-hop delays accumulate, so more routers on the path directly increases total end-to-end delay.

Q4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic. (BL4)

Answer: QoS mechanisms classify and mark traffic, then use prioritized scheduling (such as priority queuing or weighted fair queuing), traffic shaping, and bandwidth reservation (e.g., RSVP/DiffServ) to give voice and video packets preferential treatment over less time-sensitive data. This reduces the delay, jitter, and packet loss experienced by real-time streams, keeping calls and video smooth even when the network is busy.

Q5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication. (BL5)

Answer: Fiber optic cable offers far higher bandwidth, much lower signal attenuation over long distances, and complete immunity to electromagnetic interference compared to copper, enabling faster and more reliable high-speed communication over greater distances. Copper remains cheaper and simpler to terminate for short runs, but is limited in distance, bandwidth, and susceptibility to interference — making fiber the better choice for high-speed backbone and long-haul links.

Q6. Analyze how network segmentation using VLANs improves performance and security. (BL4)

Answer: VLANs logically divide a physical network into separate broadcast domains, which reduces unnecessary broadcast traffic and improves overall performance. They also isolate different departments or device types from one another, so a compromised or misbehaving host in one VLAN cannot directly reach devices in another VLAN without passing through a router/firewall, which strengthens security through segmentation.

Q7. Compare the effects of TCP and UDP on network performance for different applications. (BL4)

Answer: TCP is connection-oriented and reliable — it guarantees ordered delivery and uses flow/congestion control, which suits applications like file transfer, email, and web browsing where correctness matters, but this adds overhead and latency from acknowledgments and retransmissions. UDP is connectionless with no delivery or ordering guarantees, giving lower overhead and latency, which makes it better suited for real-time applications like VoIP, video streaming, and online gaming, where speed matters more than perfect delivery.

Q8. Why is load balancing important in large enterprise networks? Analyze its impact on performance and availability. (BL5)

Answer: Load balancing distributes traffic across multiple links or servers, preventing any single path from becoming a bottleneck and improving overall throughput and response time. It also provides redundancy — if one path or server fails, traffic is automatically shifted to the remaining resources, which significantly increases network availability and resilience for large enterprises with high traffic volumes.

Q9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization. (BL4)

Answer: Frequent routing updates allow a network to react quickly to topology changes, improving convergence time and helping traffic avoid failed or congested paths, which improves efficiency during outages. However, sending updates too frequently consumes extra bandwidth and router CPU/memory to process them, so if updates are too aggressive, the overhead itself can reduce the bandwidth available for actual user traffic.

Q10. Evaluate the trade-off between network security measures (such as firewalls and encryption) and overall network performance. (BL5)

Answer: Security measures such as firewalls and encryption require processing overhead — deep packet inspection and encrypting/decrypting traffic both add latency and consume CPU resources, which can reduce throughput. Stronger security generally comes at some performance cost, so network designers must balance the depth of inspection and strength of encryption against the acceptable latency and throughput for the application in question.

Section B: Scenario-Based / Problem-Solving Questions (Q11–Q20)

Q11. A network has high delay—how will you identify and fix the issue? (BL4)

Answer: I would first use tools like ping and traceroute to identify where along the path the delay is occurring, then check link utilization and router queue statistics to see if congestion is the cause. I'd also check for QoS misconfiguration, inefficient/indirect routing, and physical layer issues (faulty cabling or interface errors). Based on the findings, I'd apply the appropriate fix — adding bandwidth, applying QoS, correcting the routing path, or replacing faulty hardware.

Q12. If throughput is low despite high bandwidth, what could be the reasons? (BL4)

Answer: Even with high bandwidth, throughput can be low due to high round-trip time (RTT) limiting how fast a TCP window can grow, packet loss causing retransmissions, congestion at a bottleneck link somewhere on the path, a misconfigured MTU causing fragmentation, an undersized TCP window, or interference/signal issues on a wireless link.

Q13. How would you reduce packet loss in a congested network? (BL4)

Answer: I would apply QoS to prioritize critical traffic, use traffic shaping or policing to smooth out bursts, appropriately size router buffers to avoid both overflow and excessive queuing delay, add bandwidth or load-balance traffic across multiple links, and use effective congestion-control mechanisms so senders back off before the network becomes overloaded.

Q14. A VoIP call is breaking frequently—how would you troubleshoot? (BL4)

Answer: I would check jitter and packet loss statistics on the path, verify that QoS is correctly prioritizing voice traffic (e.g., DSCP marking), check for bandwidth congestion, test the Wi-Fi signal quality if wireless is involved, review the jitter buffer configuration, and use ping/traceroute to isolate which hop is introducing the problem.

Q15. Given a network with high jitter, suggest improvements (BL4)

Answer: To reduce high jitter, I would implement a jitter buffer to smooth out variable arrival times, apply QoS to prioritize time-sensitive traffic ahead of bulk data, switch to a wired connection where possible to avoid wireless variability, reduce overall network congestion, and ensure traffic consistently follows the same low-latency path rather than fluctuating routes.

Q16. How will you simulate a congested network in Packet Tracer? (BL5)

Answer: I would generate high traffic between multiple end devices (e.g., continuous pings or simulated data transfers) to load the links, deliberately reduce the bandwidth on a connecting interface to create a bottleneck, and use Packet Tracer's Simulation Mode to observe queuing behavior and packet drops, recording delay and loss statistics under this load.

Q17. If packets are delayed in a router, what steps will you take? (BL4)

Answer: I would check the router's CPU and memory utilization, review queue/buffer statistics for signs of congestion, verify there are no misconfigured ACLs adding processing overhead, check the routing table for inefficient paths or loops, and inspect interfaces for errors or duplex mismatches — then correct whichever issue is found, or upgrade hardware if the router is simply under-resourced.

Q18. How would you optimize a network for online gaming? (BL4)

Answer: I would minimize latency by using a wired connection instead of Wi-Fi, apply QoS to prioritize gaming traffic over background downloads, choose a nearby game server to reduce RTT, ensure a stable low-congestion path to reduce jitter, and disable other bandwidth-heavy applications running on the same network during play.

Q19. Design a small network and explain how you will measure its performance (BL5)

Answer: I would design a simple star-topology LAN with a router, a switch, a few PCs, and a server. To measure performance, I would use ping to measure delay and RTT, traceroute to examine the path and hop count, a file transfer to measure achievable throughput, and Packet Tracer's simulation statistics to record packet loss and jitter, comparing the results against expected baseline values.

Q20. If RTT is very high, how will you diagnose the issue? (BL4)

Answer: I would use traceroute/ping to see which specific hop introduces the largest jump in delay, check whether the distance involved makes propagation delay a natural factor, look for congestion or an overloaded router at the problem hop, and check whether the path is unnecessarily long or indirect — then address the specific cause found.

Section C: Communication & Presentation Questions (Q21–Q30)

Q21. Explain network performance to a non-technical person (BL5)

Answer: I would compare it to traffic on a highway: throughput is how many cars (data) pass a point per minute, delay is how long the trip takes from start to end, and packet loss is like cars breaking down and never completing the trip. Good network performance means data trips are fast, complete, and consistent — just like a well-managed highway with free-flowing traffic.

Q22. Describe throughput using a real-life example (BL5)

Answer: Throughput is like the actual amount of water flowing out of a pipe every second under real-world conditions — even if the pipe is rated for a maximum flow (bandwidth), the water pressure, leaks, or blockages mean the actual flow (throughput) achieved is often lower than that maximum rating.

Q23. Present how packet loss affects video streaming (BL5)

Answer: When packets are lost during video streaming, some video frames are skipped or corrupted, which shows up as pixelation, frozen frames, or stuttering playback for the viewer. To compensate, the player may buffer more heavily or reduce quality, and once loss crosses a certain threshold, the stream becomes unwatchable, directly linking network packet loss to a poor viewing experience.

Q24. Explain delay types clearly in under 1 minute (BL4)

Answer: There are four main delay types: propagation delay, the time for a signal to travel across the physical medium; transmission delay, the time needed to push all the bits of a packet onto the link; processing delay, the time a router takes to examine the packet and decide where to forward it; and queuing delay, the time the packet waits in a buffer before it can be transmitted.

Q25. Describe Cisco Packet Tracer in a professional way (BL5)

Answer: Cisco Packet Tracer is a network simulation and visualization tool that allows users to design, configure, and test realistic networking scenarios — including routers, switches, and end devices — without needing physical hardware. It is widely used in networking education to practice configuring protocols, verifying connectivity, and measuring performance metrics in a safe, repeatable environment.

Q26. Explain QoS in simple and technical terms (BL5)

Answer: In simple terms, QoS is like a priority lane that lets urgent traffic — such as voice calls or video — move ahead of less time-sensitive data. Technically, QoS classifies and marks traffic (using DSCP or CoS values), then applies scheduling techniques such as priority queuing or weighted fair queuing, along with traffic shaping and admission control, to guarantee bandwidth and limit delay, jitter, and loss for prioritized flows.

Q27. How would you present your simulation results to a client? (BL4)

Answer: I would begin with a brief summary of the objective and the key metrics measured, then use clear visuals such as graphs and tables to show trends in throughput, delay, and packet loss. I would connect these findings to what matters to the client — such as reliability or user experience — and close with clear, actionable recommendations, avoiding unnecessary technical jargon.

Q28. Explain jitter using a real-world analogy (BL5)

Answer: Jitter is like a group of friends who are supposed to arrive one after another at regular five-minute intervals, but instead show up unpredictably — some early, some late. That inconsistency in arrival timing is exactly what jitter does to data packets, and it's why voice or video calls sound choppy when jitter is high, even if no packets are actually lost.

Q29. Describe how you tested network performance in a lab (BL4)

Answer: I set up devices in Packet Tracer (and where available, a physical lab), used ping to measure round-trip delay, ran file transfers to measure achievable throughput, deliberately introduced congestion (such as high traffic load) to observe packet loss, and recorded the results in tables/graphs so I could compare performance before and after applying optimizations like QoS.

Q30. Explain the importance of performance metrics in one structured answer (BL5)

Answer: Performance metrics such as throughput, delay, jitter, and packet loss let network administrators objectively measure network health rather than relying on guesswork. They help detect problems early, plan capacity for future growth, verify compliance with service-level agreements, and confirm that configuration changes or optimizations genuinely improve the end-user experience.

Section D: HR / Behavioral Questions (Q31–Q40)

Q31. Introduce yourself and explain your knowledge of network performance (BL5)

Answer: I'm an engineering student studying computer networks, with hands-on experience using Cisco Packet Tracer to design, configure, and analyze network topologies. I have a solid understanding of key performance metrics — throughput, delay, jitter, and packet loss — and I enjoy applying these concepts to build networks that are both efficient and reliable.

Q32. Why are you interested in networking and simulation tools? (BL5)

Answer: Networking is the backbone that connects our digital world, and I find it fascinating to see how specific design choices directly affect real performance. Simulation tools like Packet Tracer let me safely experiment with different scenarios, test 'what-if' situations, and build a clear understanding of cause and effect before ever touching real hardware.

Q33. What makes you confident in analyzing network performance? (BL5)

Answer: My confidence comes from combining theoretical understanding — of concepts like congestion control, queuing, and QoS — with practical, hands-on experience simulating and measuring these metrics in Packet Tracer. Repeated practice troubleshooting simulated network issues has given me a methodical approach to diagnosing real performance problems.

Q34. Explain a concept you are very strong in (related to networking) (BL5)

Answer: I'm particularly strong in understanding congestion control and QoS. I can explain how TCP dynamically adjusts its sending rate in response to network congestion, and how QoS mechanisms prioritize traffic to protect latency-sensitive applications like voice and video, even during periods of high network load.

Q35. How would you handle a question you don't know? (BL5)

Answer: I would stay calm and honest, acknowledging that I'm not fully certain rather than guessing incorrectly. I would share my best reasoning based on related concepts I do understand, and offer to research the specific detail and follow up — I believe a structured, honest approach builds more trust than pretending to know something I don't.

Q36. Describe a situation where you solved a technical problem (BL4)

Answer: During a lab exercise, a simulated network was experiencing unexpectedly high packet loss. I systematically checked the configuration and traced the issue to a duplex mismatch between two connected interfaces. After correcting the interface settings to match, packet loss returned to normal — the experience reinforced the value of methodically isolating root causes rather than guessing at fixes.

Q37. Why should we select you for a networking role? (BL5)

Answer: I bring a combination of solid theoretical understanding of networking concepts and hands-on simulation experience, along with strong analytical and problem-solving skills. I'm genuinely interested in continuously improving, which means I can both understand and effectively troubleshoot real-world network performance issues — making me a reliable contributor to a networking team.

Q38. What challenges did you face while learning Packet Tracer? (BL5)

Answer: Initially, I found it challenging to interpret packet flows in Simulation Mode and to correctly configure routing protocols like OSPF. I overcame this through repeated practice, following structured tutorials step by step, and deliberately breaking and then fixing configurations, which helped me build a much stronger intuition for how the underlying protocols actually behave.

Q39. How do you improve your technical skills regularly? (BL5)

Answer: I regularly build and test new network topologies in Packet Tracer to reinforce concepts, follow networking blogs and official documentation to stay current, take on small self-directed projects, and make a point of revisiting any concept I find difficult until I can explain it simply and confidently to someone else.

Q40. Demonstrate confidence while explaining a complex concept simply (BL5)

Answer: Take TCP congestion control as an example: think of it like a driver in traffic who gradually speeds up when the road is clear, but brakes sharply the moment they see brake lights ahead. In the same way, TCP steadily increases its sending rate until it detects congestion (packet loss), then cuts back quickly before ramping up again — constantly balancing speed against the risk of overloading the network.